



## **Moving towards a fairer hiring process**

By: Kina Kim, Fatima Masood, Shuting Xie, Iman Cheema, Aaina Garg

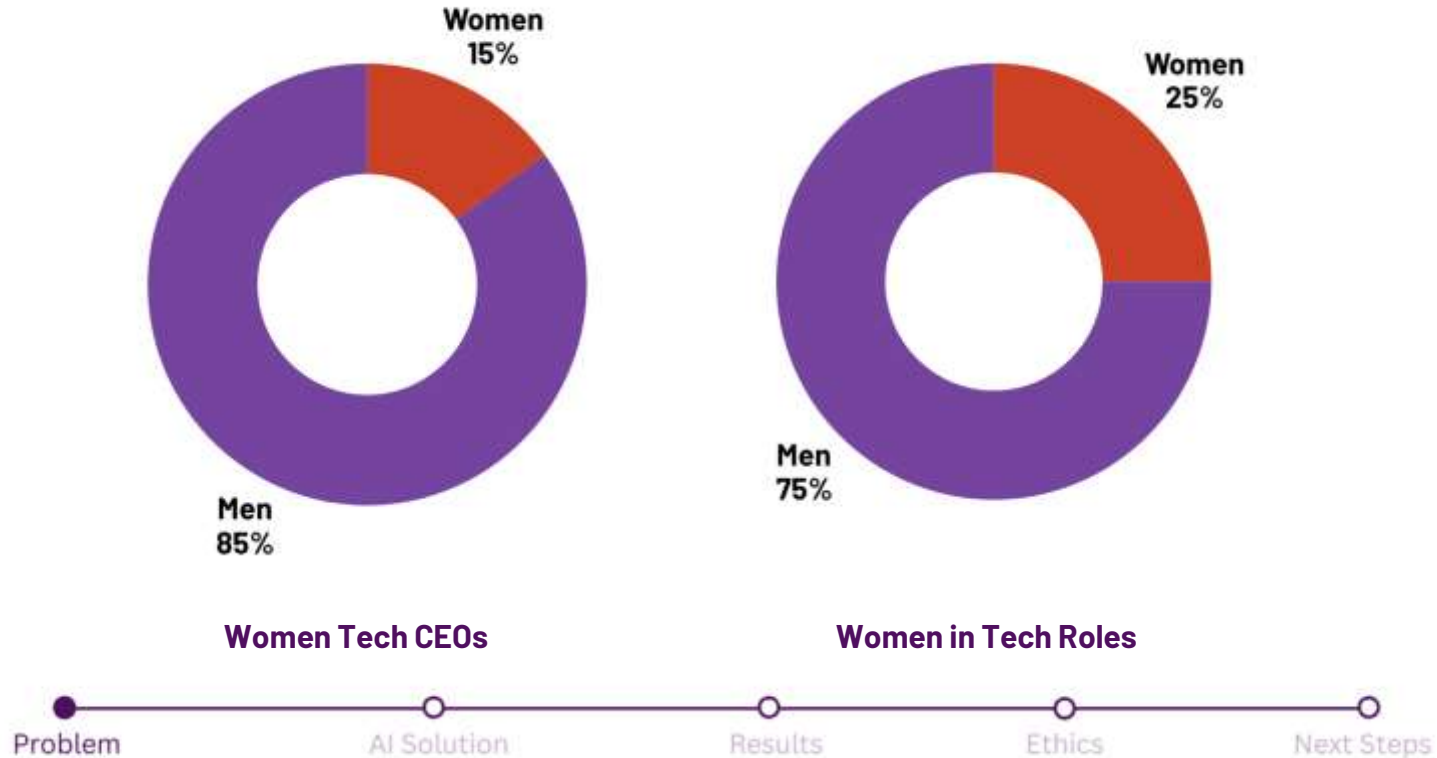
# Meet McKelvey

*Ignored as Erin. Hired as Mack.*

*\*based on true story*



# She's Not Alone



# Amazon Story



Machine  
Learning  
Department

Carnegie Mellon University  
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October 11, 2018

Amazon Scraps Secret AI Recruiting Engine that Showed Biases Against Women

Top U.S. tech companies have yet to close the gender gap in hiring, a disparity most pronounced among technical staff such as software developers where men far outnumber women. Amazon's experimental recruiting engine followed the same pattern, **learning to penalize resumes including the word "women's"** until the company discovered the problem.

Problem

AI Solution

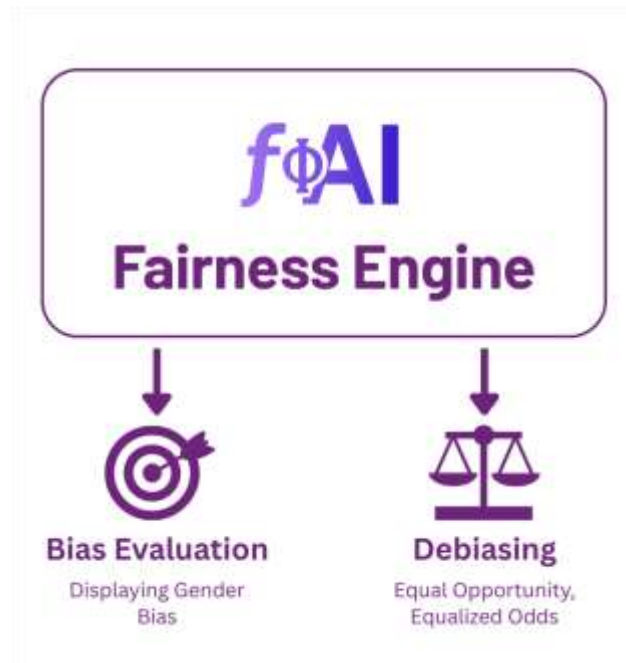
Results

Ethics

Next Steps

# Introducing fAI

AI hiring tools often **reinforce existing biases** by learning from unfair historical data.

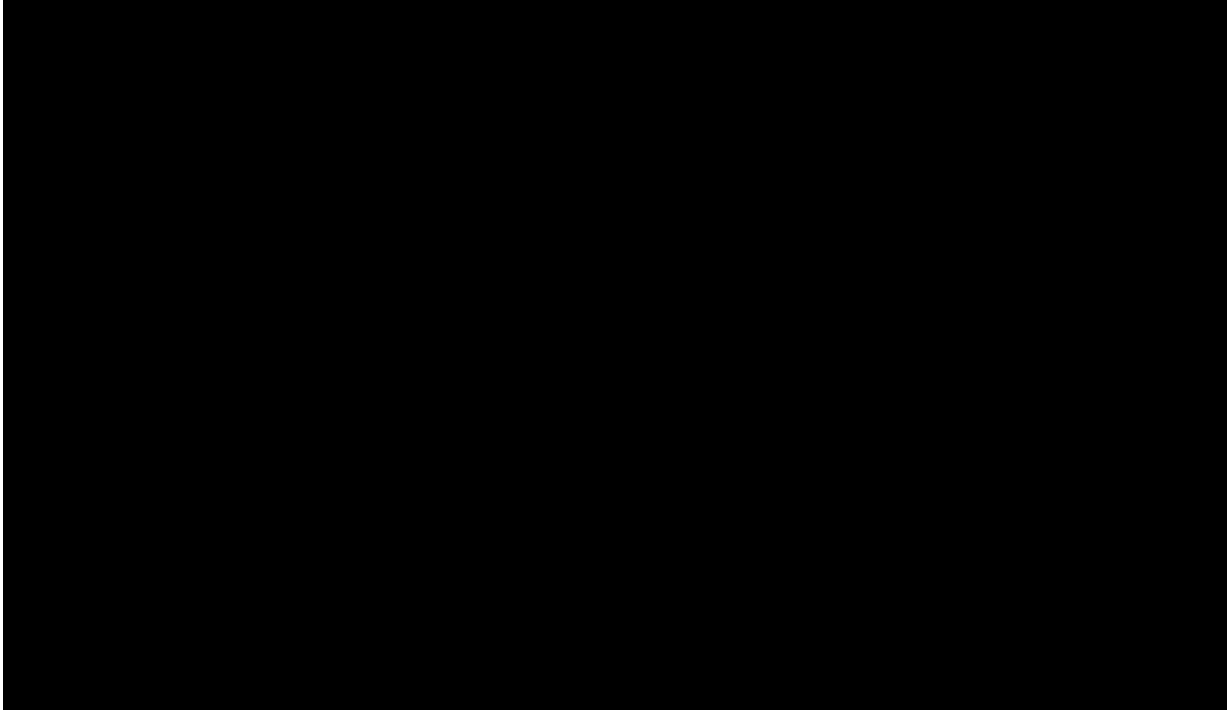




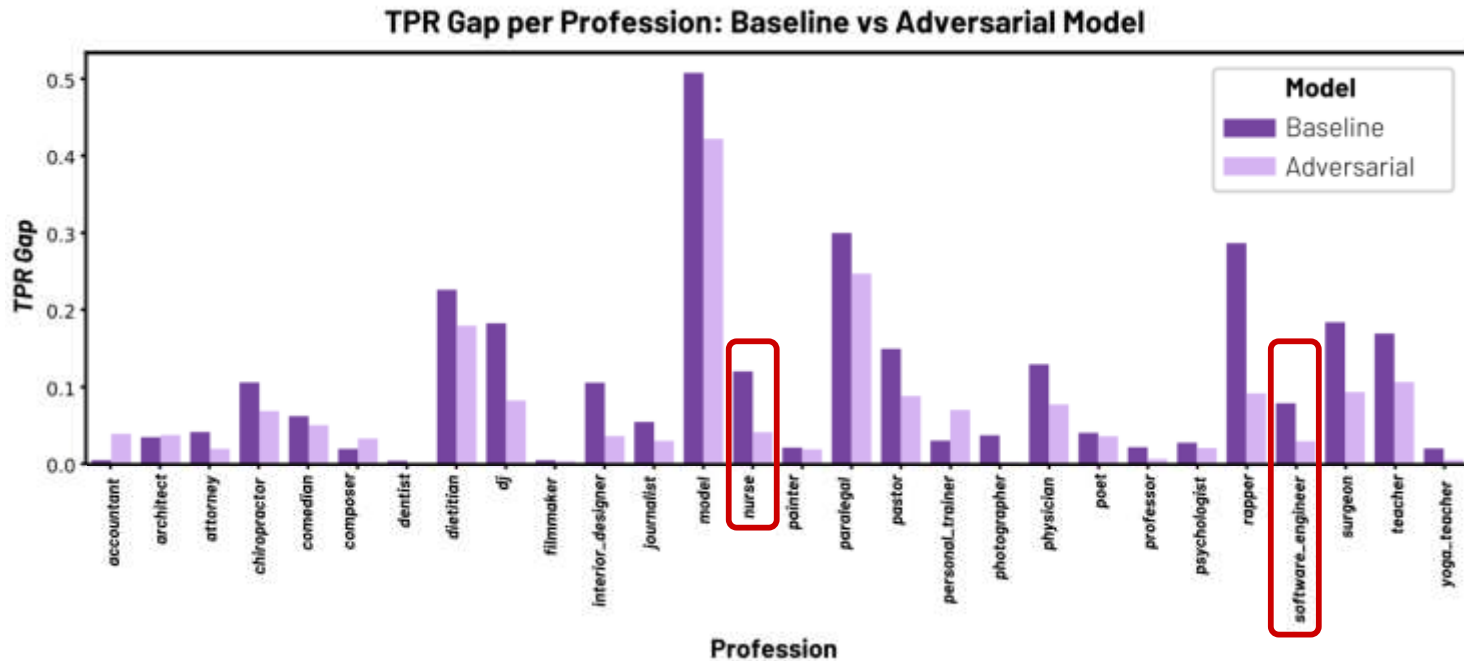
## Training **without** the Adversary



# User Experience



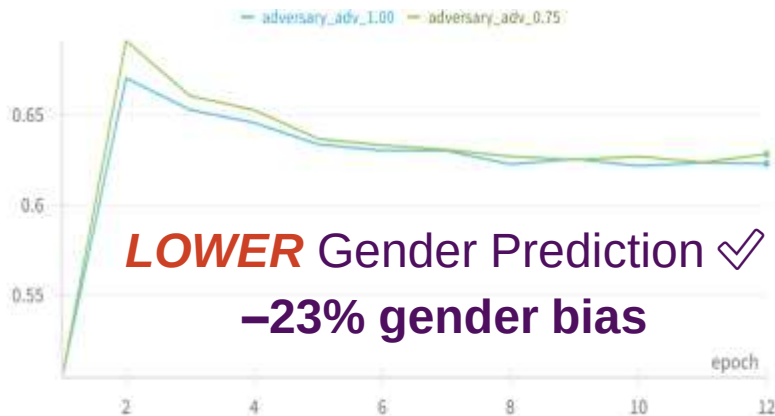
# Results: Bias Reduction





# Results: Accurate yet Fair

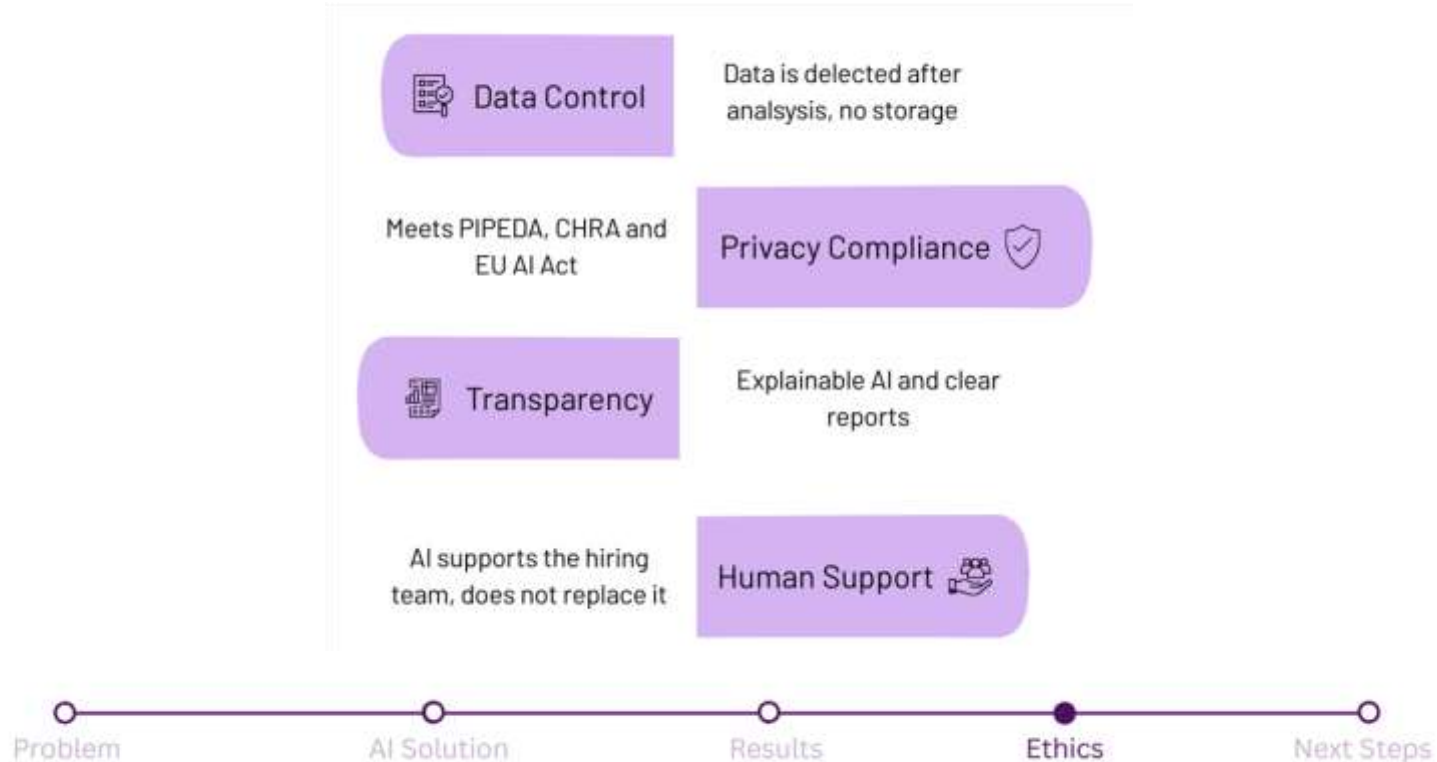
### Validation Adversary Accuracy



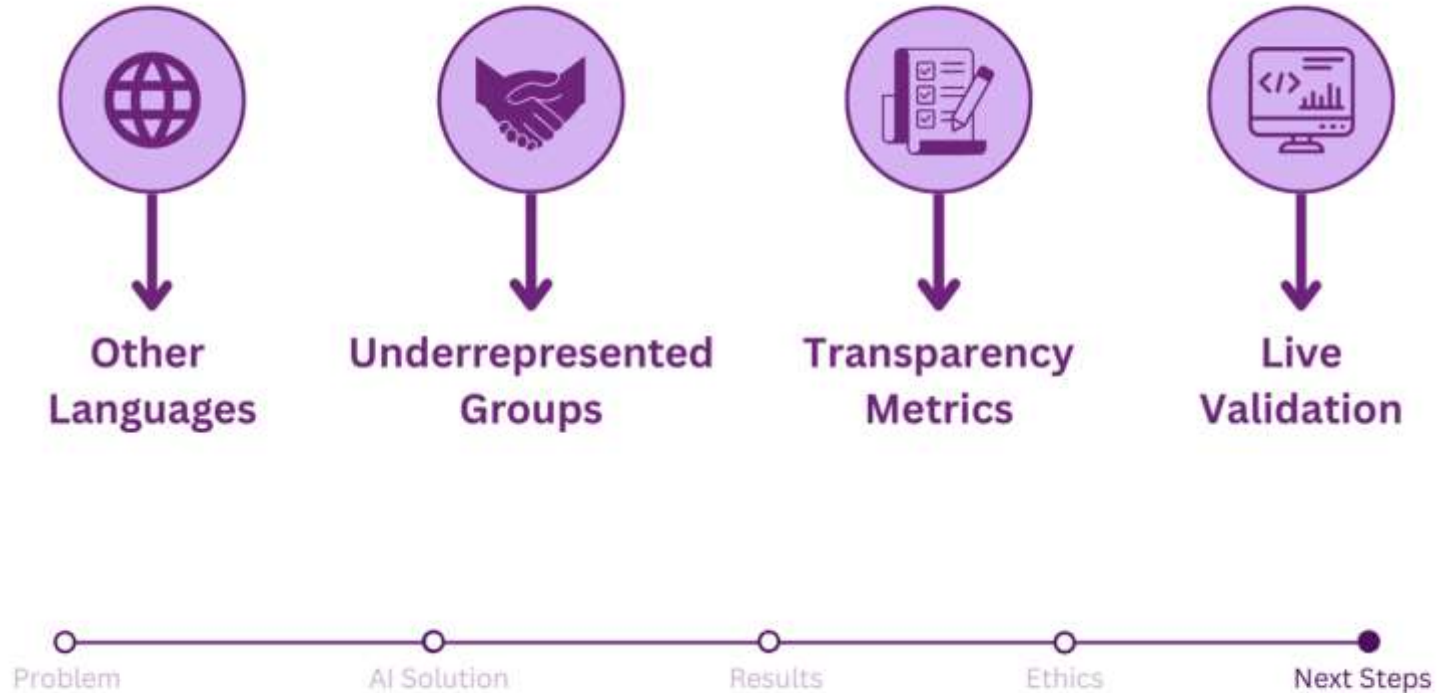
### Validation Classifier Accuracy



# Ethical Considerations



# Future Work





Kina (Gi Yeon) Kim  
Machine Learning Engineer



Fatima Masood  
Machine Learning Engineer



Shuting Xie  
Machine Learning Engineer



Iman Cheema  
Front + Backend Engineer



Aaina Garg  
Front + Machine Learning Engineer

*Because FAIR hiring matters*

# Thank You



**Sedef Koçak**  
Industry Mentor at Vector



**Mia Thomas**  
Academic Mentor (UofT)



AI4GOOD Lab



Vector Institute



CIFAR

# Thank You

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# References

Zhang, B. H., Lemoine, B., & Mitchell, M. (2018). Mitigating unwanted biases with adversarial learning. Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society, 335–340. <https://doi.org/10.1145/3278721.3278779>



**Moving towards a fairer hiring process**

Come find us at our booth



# APPENDIX

# A1. MODEL

# i. Inspiration

## Mitigating Unwanted Biases with Adversarial Learning

**Brian Hu Zhang**  
Stanford University  
Stanford, CA  
bhz@stanford.edu

**Blake Lemoine**  
Google  
Mountain View, CA  
lemoine@google.com

**Margaret Mitchell**  
Google  
Mountain View, CA  
mmitchellai@google.com

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## Achieving Fairness through Adversarial Learning: an Application to Recidivism Prediction

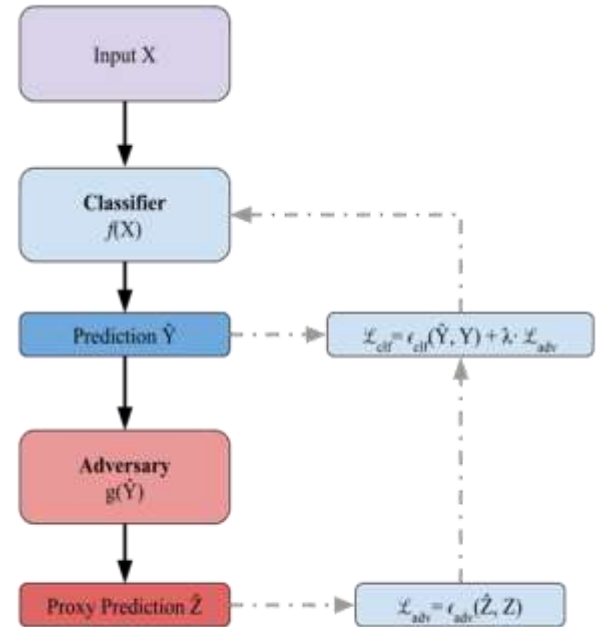
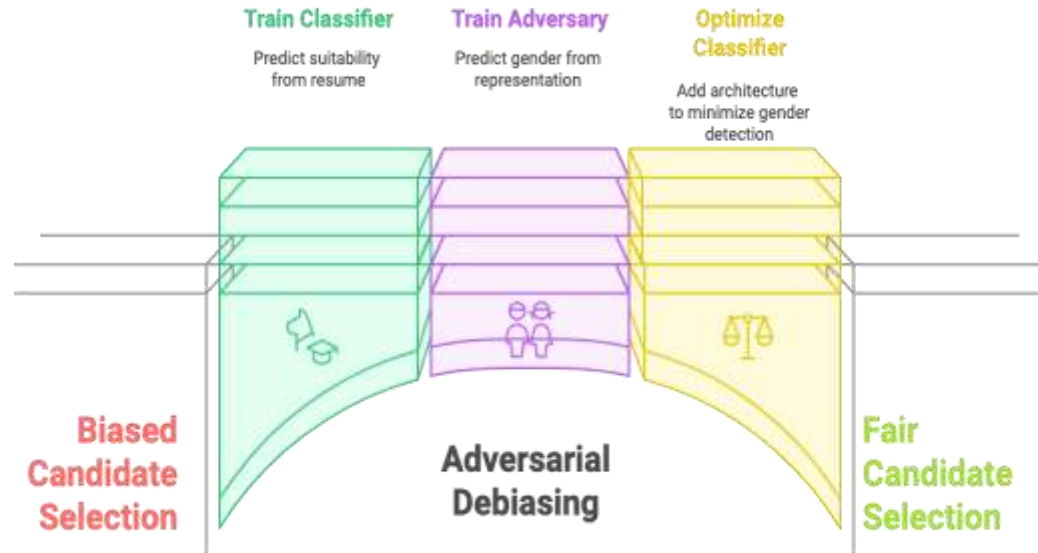
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**Christina Wadsworth**  
Stanford University  
Stanford, CA  
cwads@cs.stanford.edu

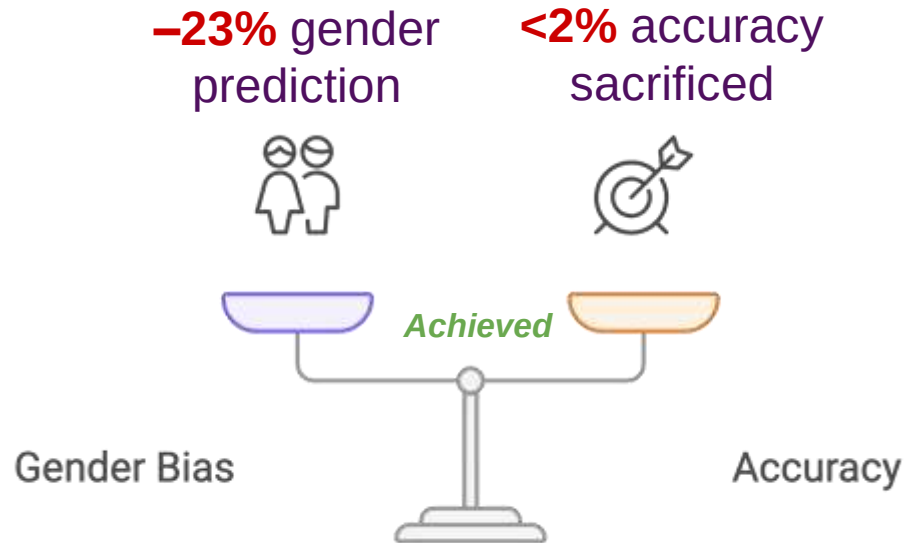
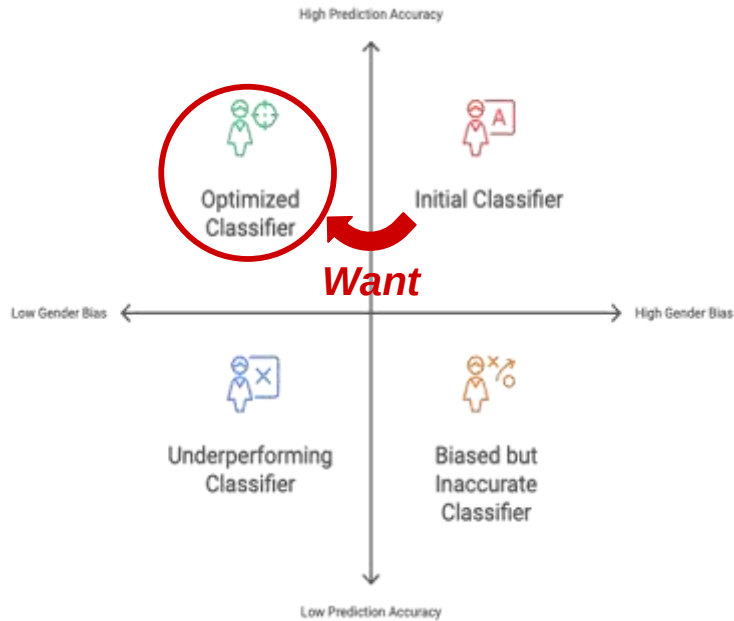
**Francesca Vera**  
Stanford University  
Stanford, CA  
fvera@cs.stanford.edu

**Chris Piech**  
Stanford University  
Stanford, CA  
piech@cs.stanford.edu

## ii. Technical Details



### iii. Objective & Achievement



## iv. Model Selection

| Model         | Baseline Performance   | Model Size    | Training Time | Inference Speed | Ease of Fine-Tuning | General Observations                               |
|---------------|------------------------|---------------|---------------|-----------------|---------------------|----------------------------------------------------|
| BiGRU         | Above Average (86-87%) | Small (~10M)  | Moderate      | Fast            | Easy                | Good balance of speed and performance              |
| BiLSTM        | High (87-88%)          | Medium (~66M) | Moderate      | Moderate        | Moderate            | Good accuracy but slightly more resource-intensive |
| DistilBERT    | Very High (89-90%)     | Large (~125M) | Slow          | Slow            | Difficult           | Best performance but slower and resource-heavy     |
| Electra-small | Moderate (84-85%)      | Small (~14M)  | Fast          | Fast            | Easy                | Efficient but not the highest performance          |

## A2. METRICS & RESULT

# i. Metrics Definition

|        |   | Prediction          |                     |
|--------|---|---------------------|---------------------|
|        |   | 0                   | 1                   |
| Actual | 0 | True Negative (TN)  | False Positive (FP) |
|        | 1 | False Negative (FN) | True Positive (TP)  |

## 1. Equal Opportunity :

The rate of **qualified** candidates being **correctly classified** into the occupation should be **the same** across gender groups.

$$TPR = \frac{TP}{TP + FN}$$

$$TPR_0 = TPR_1 \quad (1)$$

$$TPR_1 - TPR_0 < \text{Cutoff} \quad (2)$$

$$\frac{TPR_0}{TPR_1} > \text{Cutoff} \quad (3)$$

## 2. Equalize Odd:

**both** qualified and unqualified candidates should be **treated similarly** across gender in terms of **correct** and **incorrect** predictions.

$$FPR = \frac{FP}{FP + TN}$$

$$TPR_0 = TPR_1$$

$$FPR_0 = FPR_1$$

## 3. Disparity Parity

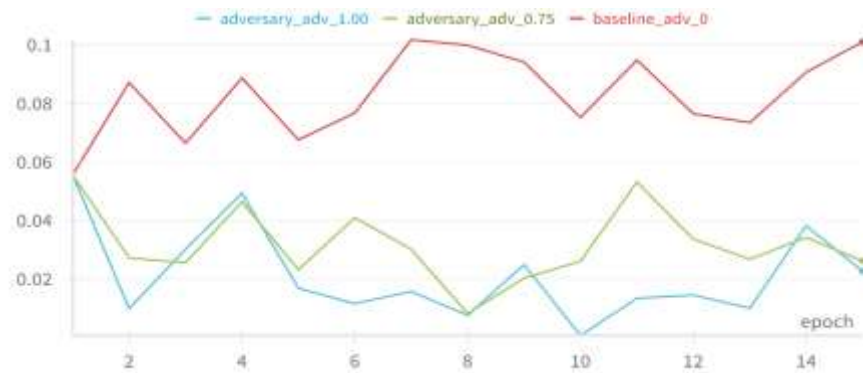


## ii. Result, Training Graph per Profession

Equal Opportunity of "Surgeon"

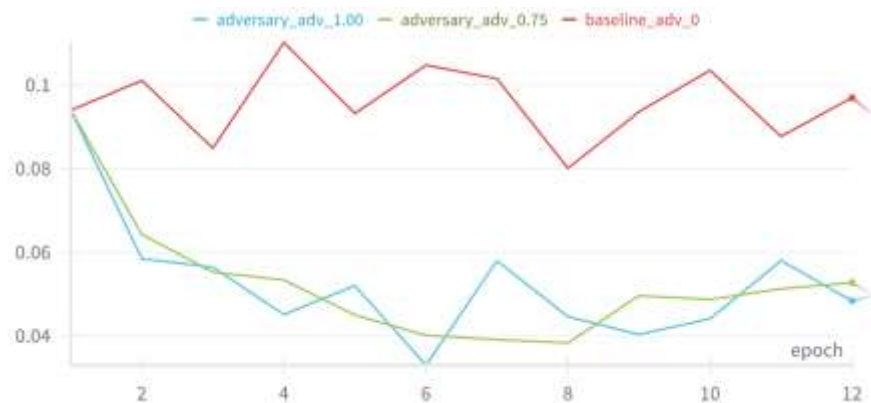


Equal Opportunity of "Nurse"

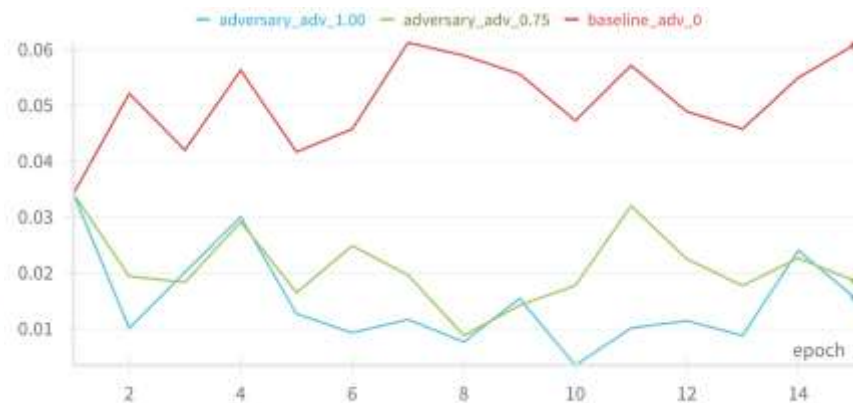


### iii. Extra Metric, Equalized Odds Metrics

Equalized Odd of "Surgeon"



Equalized Odd of "Nurse"



## A3. IMPACT

## i. Limitation in ML Screener

Amazon's recruiting machine learning model was trained to vet applicants by analyzing certain parameters in resumes submitted to the company over a 10-year period. **Due to the biases that the machine learning model had, most ideal candidates were generated as men**, which is a reflection of the male dominance across the tech industry — therefore the data fed to the model was not unbiased towards gender equality but au contraire.

Amazon's research team states that they modified the central algorithms and made the machine learning model neutral to these gender biases, however that was not a guarantee that the engine would not devise other ways of sorting candidates (**i.e. male dominant keywords in applicant's resumes**) that could prove discriminatory.

## ii. Criticality

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### Women In Tech Stats: How The Industry Can Provide Equal Opportunities

By Cheryl Robison, Contributor  Dr. Cheryl Robison covers areas of lead... [Follow Author](#)

Jul 16, 2024, 05:00pm EDT

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arXiv:2405.19699v3 [cs.CY] 18 May 2025

## Fairness in AI-Driven Recruitment: Challenges, Metrics, Methods, and Future Directions

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### AI hiring tools may be filtering out the best job applicants

16 February 2024

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#### TECH

**Amazon reportedly scraps internal AI recruiting tool that was biased against women** / The secret program penalized applications that contained the word “women’s”

by [James Vincent](#)  
Oct 10, 2018, 7:09 AM EDT

### Ethics and discrimination in artificial intelligence-enabled recruitment practices

[Zhisheng Chen](#) 

[Humanities and Social Sciences Communications](#) **10**, Article number: 567 (2023) | [Cite this article](#)

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  **PRINCETON UNIVERSITY**

## Biased bots: Artificial-intelligence systems echo human prejudices

## A4. INNOVATION

## i. Insufficiency with masking

### **Bias in Bios: A Case Study of Semantic Representation Bias in a High-Stakes Setting**

Maria De-Arteaga<sup>1</sup>, Alexey Romanov<sup>2</sup>, Hanna Wallach<sup>3</sup>, Jennifer Chayes<sup>3</sup>, Christian Borgs<sup>3</sup>,  
Alexandra Chouldechova<sup>1</sup>, Sahin Geyik<sup>4</sup>, Krishnaram Kenthapadi<sup>4</sup>, Adam Tauman Kalai<sup>3</sup>

<sup>1</sup>*Carnegie Mellon University*, <sup>2</sup>*University of Massachusetts Lowell*, <sup>3</sup>*Microsoft Research*, <sup>4</sup>*LinkedIn*

online biographies other than explicit gender indicators. These differences may be due to the ways that men and women represent themselves or due to men and women having different specializations within an occupation. **Our findings highlight both the risks of using machine learning in a high-stakes setting and the difficulty of trying to promote fairness by “scrubbing” sensitive attributes.**

## ii. Word Embedding (choice of word) contains bias

Man is to Computer Programmer as Woman is to Homemaker?

### Debiasing Word Embeddings

Tolga Bolukbasi<sup>1</sup>, Kai-Wei Chang<sup>2</sup>, James Zou<sup>2</sup>, Venkatesh Saligrama<sup>1,2</sup>, Adam Kalai<sup>2</sup>

<sup>1</sup>Boston University, 8 Saint Mary's Street, Boston, MA

<sup>2</sup>Microsoft Research New England, 1 Memorial Drive, Cambridge, MA

tolgab@bu.edu, kw@kwchang.net, jamesyzou@gmail.com, srv@bu.edu, adam.kalai@microsoft.com

#### Abstract

The blind application of machine learning runs the risk of amplifying biases present in data. Such a danger is facing us with *word embedding*, a popular framework to represent text data as vectors which has been used in many machine learning and natural language processing tasks. We show that even word embeddings trained on Google News articles exhibit female/male gender stereotypes to a disturbing extent. This raises concerns because their widespread use, as we describe, often tends to amplify these biases. Geometrically, gender bias is first shown to be captured by a direction in the word embedding. Second, gender neutral words are shown to be linearly separable from gender definition words in the word embedding. Using these properties, we provide a methodology for modifying an embedding to remove gender stereotypes, such as the association between the words *receptionist* and *female*, while maintaining desired associations such as between the words *queen* and *female*. We define metrics to quantify both direct and indirect gender biases in embeddings, and develop algorithms to “debias” the embedding. Using crowd-worker evaluation as well as standard benchmarks, we empirically demonstrate that our algorithms significantly reduce gender bias in embeddings while preserving its useful properties such as the ability to cluster related concepts and to solve analogy tasks. The resulting embeddings can be used in applications without amplifying gender bias.



## ii. Word Embedding (choice of word) contains bias

### Gender stereotype *she-he* analogies.

|                     |                             |                           |
|---------------------|-----------------------------|---------------------------|
| sewing-carpentry    | register-nurse-physician    | housewife-shopkeeper      |
| nurse-surgeon       | interior designer-architect | softball-baseball         |
| blond-burly         | feminism-conservatism       | cosmetics-pharmaceuticals |
| giggle-chuckle      | vocalist-guitarist          | petite-lanky              |
| sassy-snappy        | diva-superstar              | charming-affable          |
| volleyball-football | cupcakes-pizzas             | hairdresser-barber        |

### Gender appropriate *she-he* analogies.

|                 |                                |                   |
|-----------------|--------------------------------|-------------------|
| queen-king      | sister-brother                 | mother-father     |
| waitress-waiter | ovarian cancer-prostate cancer | convent-monastery |

## ii. Word Embedding (choice of word) contains bias

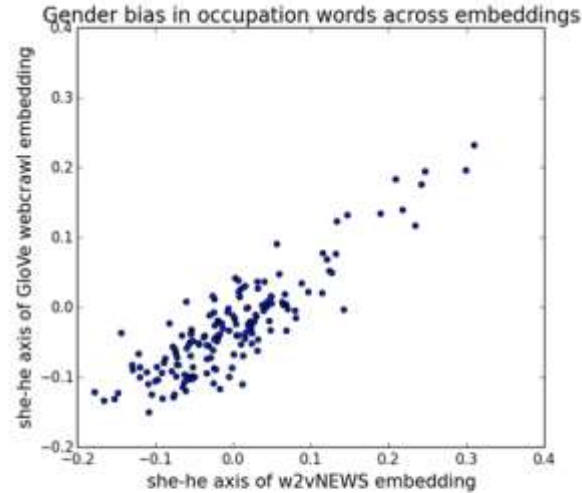


Figure 4: Comparing the bias of two different embeddings—the w2vNEWS and the GloVe web-crawl embedding. In each embedding, the occupation words are projected onto the *she-he* direction. Each dot corresponds to one occupation word; the gender bias of occupations is highly consistent across embeddings (Spearman  $\rho = 0.81$ ).

### iii. Future, Individual Fairness

#### Group Fairness :

"On average, X% of each group should be treated the same way"

#### Earthmover Distance

the minimum "effort" to transform one distribution into the other



#### Individual Fairness :

"Similar Individuals should receive similar outcomes"



#### Lipschitz Condition

$$|f(x_1) - f(x_2)| \leq K \cdot d(x_1, x_2)$$

If K is **small**, this **guarantees** individual fairness



## iv. Other Applications

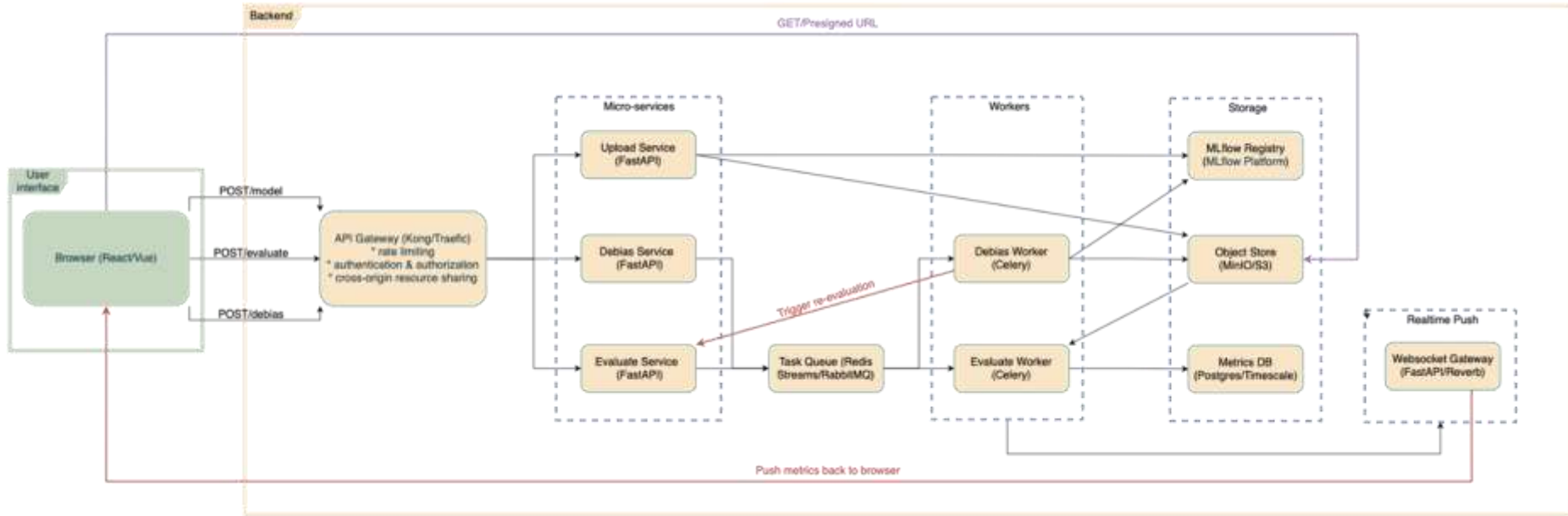
Any application in ML used for **resource allocation**  
has potential to adopt this framework

Example,

1. Loan Approval / Credit Scoring
2. University Admission / Scholarship Allocation
3. Criminal Justice / Recidivism Prediction

# A5. WEBSITE

## i. Website Architecture



# A6. ETHICS

## i. Ethical Principles in the context of Fair AI



**Reduces  
gender bias**



**Increases  
system clarity**



**Maintains  
human oversight**



**Ensures  
responsible use**



**Improves  
system security**



**Promotes  
diverse representation**



## ii. PIPEDA (Personal Information Protection and Electronic Documents Act, Canada)

1. Consent (Principle 3): Must obtain meaningful consent to collect, use, or disclose personal information. Individuals must know the purpose of data collection and use
2. Limiting Collection, Use, and Retention (Principles 4 & 5): Only collect and keep data necessary for stated purposes, and delete it when no longer needed
3. Openness & Transparency (Principle 8): Organizations must be open about their policies and practices regarding the management of personal information
4. Individual Access (Principle 9): Individuals have the right to access their personal data and know how it's used
5. Accuracy & Safeguards (Principles 6 & 7): Data must be accurate and protected by appropriate security measures
6. Challenging Compliance (Principle 10): Individuals must be able to challenge an organization's compliance with these principles

### iii. CHRA (Canadian Human Rights Act)

1. Prohibited Grounds for Discrimination (Section 3): Employment decisions must not discriminate on grounds such as race, national/ethnic origin, color, religion, age, sex, sexual orientation, gender identity/expression, marital status, family status, disability, or conviction for which a pardon has been granted.
2. Duty to Accommodate: Employers must accommodate individuals up to the point of undue hardship, including in hiring and assessment processes.
3. Harassment Prohibition: Protection from harassment based on prohibited grounds

## iv. EU AI Act

- High-Risk Classification (Article 6(2), Annex III): AI used for recruitment, selection, or employment decisions is classified as “high-risk” and subject to strict requirements.
- Transparency (Article 13): Must provide clear information about the AI system’s capabilities, limitations, and logic to users and affected individuals.
- Human Oversight (Article 14): High-risk AI must allow for effective human oversight—decisions should not be fully automated without human review.
- Data Governance (Article 10): Requires high-quality, representative, and bias-mitigated datasets.
- Record-Keeping (Article 12): Providers must keep detailed logs and documentation of system operation and decisions.
- Risk Management (Article 9): Ongoing risk assessment and mitigation for discriminatory or harmful outcomes.
- Prohibited Practices (Article 5): Bans certain uses, such as emotion recognition in the workplace and manipulative or exploitative AI

# Feasibility (Market Research)

# Market Growth Projections

## Growth Projections in HR Technology Market

### North American HR Tech Market

The North American HR tech market is estimated at \$24B in 2025, with over 10,000 enterprise clients seeking compliance and fairness in hiring.

### Global HR Tech Market

The global HR technology market is projected to reach \$84.7B by 2032 (CAGR: 9.3%), driven by rapid digital transformation, cloud adoption, and regulatory demands for ethical AI.

### Global AI Recruitment Tools Market

The global AI recruitment tools market is expected to triple from \$1B in 2023 to \$3B by 2032 (CAGR: 12.5%).

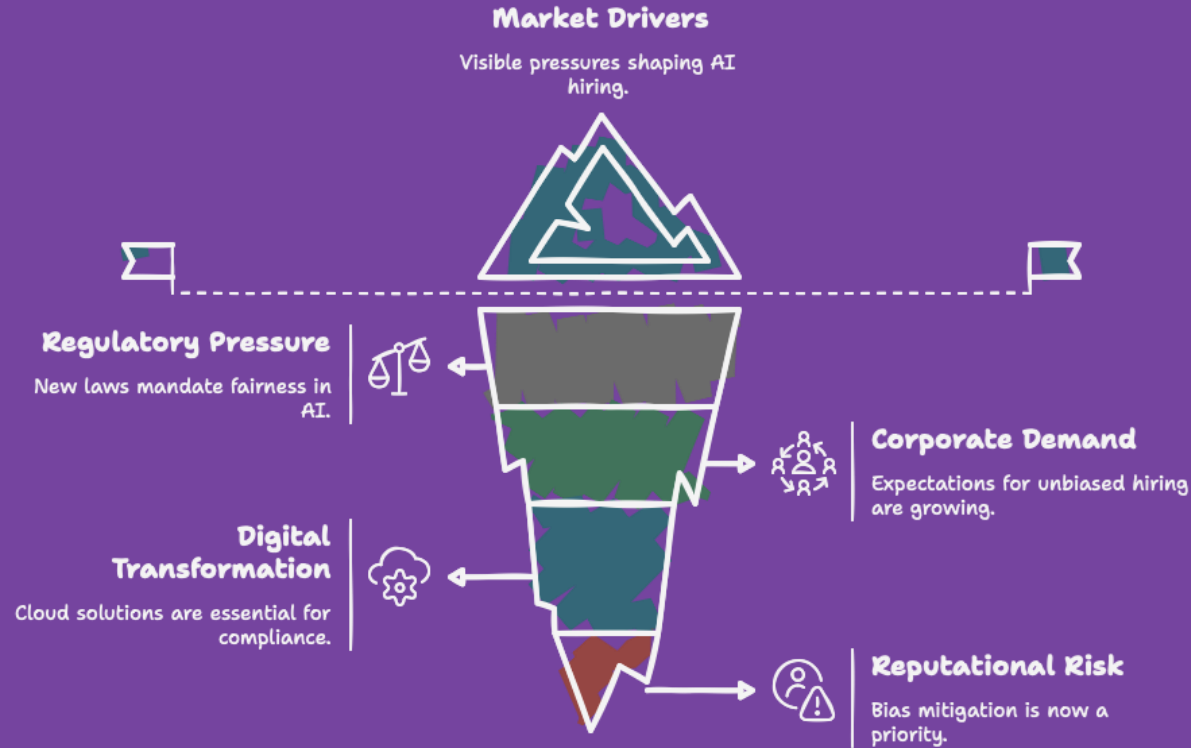
### AI Fairness Segment

The AI fairness segment alone is forecasted to reach \$2.5B by 2025 (CAGR: 28%).



# Market Drivers

# AI Hiring: Unveiling the Hidden Depths





# Target Customers

# TARGET CUSTOMERS



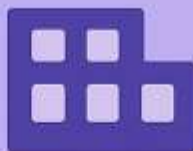
## **ENTERPRISE HR DEPARTMENTS & RECRUITMENT AGENCIES**

Large organizations  
with DEI mandates  
in tech, finance,  
public sectors

## **RECRUITMENT PROCESS OUTSOURCING (RPO) PROVIDERS**



Firms managing hiring for others  
needing unbiased, data-driven processes



## **MID-SIZED COMPANIES & SMEs**

Digitizing HR-  
seeking plug-and-lay  
AI for fair hiring



## **IDEAL CLIENT PROFILE**

- Companies with potential biases in hiring across any sector
- Organizations under regulatory audit or public scrutiny
- HR teams seeking automation with human oversight

# FAIRAI vs Legacy Tools

## ii. Competitive USP

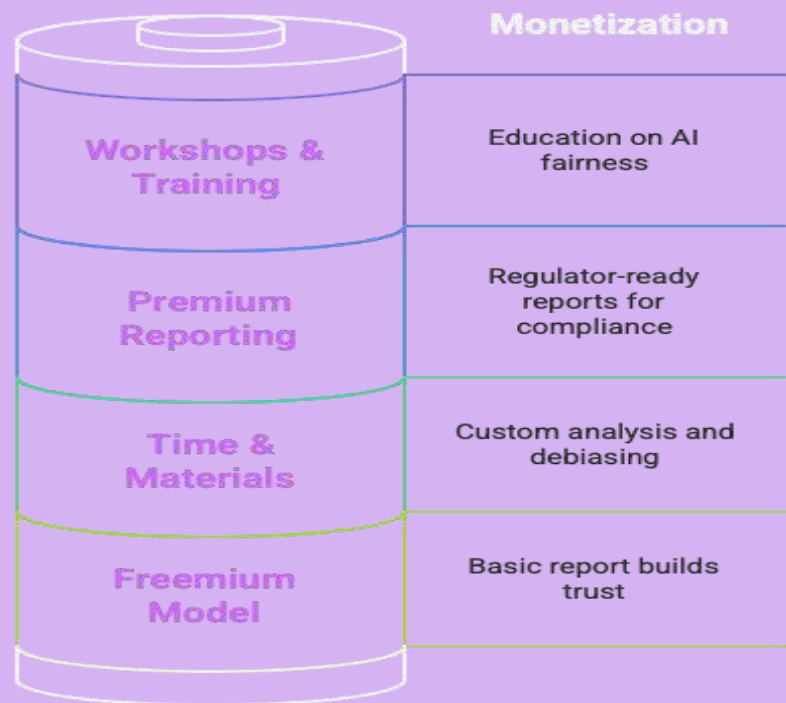
1. Privacy-first, **MODEL-ONLY** APPROACH
2. Regulatory-ready Documentation and Compliance
3. Plug and Play with any model
4. Transparent, Flexible pricing
5. Continuous monitoring and Model Governance
6. Education and Enablement

## Choose Fair AI for Equitable Hiring Practices



# Monetization Next Steps

**Monetization strategies range from free to highly customized services.**



## A6. SAMPLE — REVENUE MODEL

### REVENUE MODEL



#### Debiasing-as-a-Service

\$1,000/month for up to 500 profiles  
\$2,500/month for up to 2,500 profiles



#### Collaborative partnership

\$4,000/month  
workshops, dashboards, support



#### Industry benchmarking & insights

\$7,500/year  
reports, strategy session